

Basic Science

Measured loads on a vertebral body replacement during sitting

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Abstract

BACKGROUND CONTEXT: Sitting is frequently assumed to cause high spinal loads because people with sedentary work often suffer from low back pain. It is assumed that the posture while sitting, as well as several seat parameters, also affects the spinal loads.

PURPOSE: To measure the loads on a spinal implant for different upper body inclinations, backrest declinations, seat heights, types of seat, and arm positions.

STUDY DESIGN: Loads on a vertebral body replacement during sitting were measured in five patients with telemeterized implants.

METHODS: The telemeterized vertebral body replacement measures all six load components. It was implanted into five patients suffering from compression fractures of a lumbar vertebral body. Loads were measured when the patients were sitting on a stool and inclining their upper body between 15° flexion and 10° extension in steps of 5°; on a chair with an adjustable backrest that allowed declination angles between 108° and 180°; on an office chair while the seat height was varied between 40 and 60 cm in steps of 5 cm; and successively on seven different types of seats. The effect of the arm position was also studied.

RESULTS: The resultant implant force was increased on the average by 48% for 15° flexion and decreased by 19% for 10° extension of the trunk. When sitting on a chair with an adjustable backrest, the loads decreased with an increasing backrest declination angle. The seat height had in most cases only a minor effect on implant loads. In comparison to sitting on a stool, the loads were reduced when sitting on a bench (7%) or a stool with a padded wedge (9%), a knee stool (19%), a chair (35%), and an office chair (41%). Sitting on a physiotherapy ball increased the loads by 7%. Placing the hands on the thighs reduced the implant loads on the average by 19% in comparison to arms hanging on the sides.

CONCLUSION: Spinal loads can be reduced by leaning against the backrest, placing the arms on the armrest or the thighs, and by decreasing the flexion angle of the upper body. © 2011 Elsevier Inc. All rights reserved.

Keywords:

Sitting; Load measurement; Vertebral body replacement; Telemetry; Spinal loads

Introduction

High spinal loads and keeping a constrained posture over a long period are reasons for back pain [1,2]. People with sedentary work often suffer from low back pain.

Sitting is frequently assumed to cause high spinal loads. However, sitting can vary by several factors, such as type of seat (eg, chair, physiotherapy ball, and knee stool), posture (eg, relaxed, erect, upper body inclined, and leaned against the backrest), and the positioning of the arms (eg, on the thighs, armrests, a table, and hanging them laterally). In addition, the inclination of the pelvis and thus the lordosis of the lumbar spine may differ strongly during sitting. Some types of seats assist certain pelvis inclinations, which are alleged to reduce spinal load or are considered to prevent back pain. These types include, for example, physiotherapy balls, modern office chairs with a tiltable backrest that permit changing the shape of the back [3], and chairs with alternating microrotatory movements of the seat [4].

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A direct complete measurement of the spinal load has not yet been possible. Nachemson [5,6] measured the intradiscal pressure and found a 40% higher pressure for sitting on a stool than for standing. Wilke et al. [7,8] measured a 10% lower disc pressure during sitting on a stool compared with standing. The loads on internal spinal fixation devices were measured by Rohlmann et al. [9,10]. They also found on the average 10% lower loads for sitting on a stool than for standing. The forces on an instrumented vertebral body replacement (VBR) for sitting on a stool and standing were compared by Dreischarf et al. [11]. They found that the load ratio “sitting to standing” depends on the curvature of the thoracic spine. Persons with a strong kyphotic thoracic spine had higher relative forces during sitting than did those with a small kyphosis. The arm position also strongly influenced the spinal load. When the arms were hanging laterally, the loads were higher than when they were placed on the thighs. In agreement with Nachemson [5,6], Dreischarf et al. [11] measured on the average a 40% higher load for sitting than for standing.

The effect of different types of seats on the loads on internal spinal fixation device was measured by Rohlmann et al. [12]. There were only minor differences in fixator loads for sitting on different types of seats. However, in that study, the patients were not allowed to lean against a backrest. Using stadiometry, Althoff et al. [13] measured the changes in body height when sitting on different types of seats. They found large load differences between the various types of seats and in all cases lower spinal loads than for standing.

The aim of the present study was to determine the effects of several parameters on the implant load during sitting. In particular, the influences of upper body inclination and upper body declination with and without leaning against a backrest, seat height, and type of seat were to be determined.

Materials and methods

Instrumented implant and patients

Six load sensors and a telemetry unit were integrated into the clinically proven implant Synex (Synthes Inc., Bettlach, Switzerland) to measure the loads (Fig. 1). The inductively powered implant allows the measurement of all three force and three moment components. Typically, average measuring errors are below 2% for force and 5% for moment components related to the maximum calibration values of 3,000 N and 20 Nm, respectively. The implant has been described in detail elsewhere [14].

Telemeterized VBRs were implanted into five patients. The Table provides data on patients and surgical procedures. Four patients (WP1–WP4) had an A3 type compression fracture (classification after Magerl et al. [15]) of the vertebral body L1 and one patient (WP5) that of the vertebral body L3. The patients were first treated with pedicle screw-based internal spinal fixation devices implanted from the posterior.

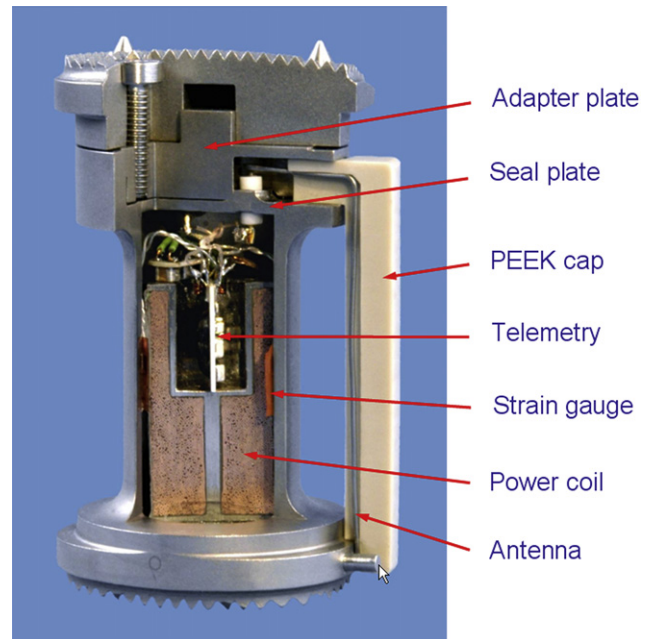


Fig. 1. Cut model of the telemeterized vertebral body replacement. (Reprinted with permission from Rohlmann A, Graichen F, Bender A, et al. Loads on a telemeterized vertebral body replacement measured in three patients within the first postoperative month. Clin Biomech (Bristol, Avon) 2008;23:147–58 [17]).

In a second surgery, parts of the fractured vertebral body and the adjacent discs were removed, and the telemeterized VBR was inserted into the corpectomy defect. Autologous bone material was added to the VBR to enhance fusion of the adjacent segments.

For an inductive power supply during the measurements, a coil was placed around the patient's trunk at the level of the implant. A loop antenna on the patient's back received the signals of the telemetry. The patients were videotaped during the measurements, and the load-dependent signals were stored on the same videotape. Forces and moments were calculated from the telemetric signals with the help of a notebook, and the loading curves were shown online on its monitor. The telemetry and the external equipment have been described in detail elsewhere [16].

Table
Data on patients and surgical procedures

Parameter	Patient				
	WP1	WP2	WP3	WP4	WP5
Gender	Male	Male	Female	Male	Male
Age at the time of surgery (y)	62	71	69	63	66
Height (cm)	168	169	168	170	180
Body mass (kg)	66	74	64	60	63
Fractured vertebra	L1	L1	L1	L1	L3
Level of internal fixation device	T12–L2	T12–L2	T11–L3	T12–L2	L2–L4
Bone material added	Yes	Yes	Yes	Yes	Yes
Implantation date (mo/y)	09/2006	11/2006	03/2007	01/2008	07/2008

The Ethics Committee of our hospital approved implantation of the modified implant in patients. Before surgery, the procedure was explained to the patients, and they gave their written consent to implantation of the telemeterized VBR, taking measurements, and publishing of their images.

Seats and exercises

Four of the five patients agreed to measurements being made with the upper body slightly inclined (between 15° flexion and 10° extension, in steps of 5°) while sitting on a stool. The patient's back inclination was measured with an inclinometer.

In addition, all five patients sat in a chair with an adjustable backrest, which allowed supported upper body declinations between 108° and 180°. The leg-supporting part of this chair moved simultaneously with the backrest. Implant loads were measured for upper body declinations between sitting upright (90°) and lying in a supine position (180°). For the upright position (90°), the patients' back was not in contact with the backrest. The effect of seat height on implant loads was measured using an office chair without a backrest, the height of which could be varied between 40 cm and 60 cm in steps of 5 cm.

For studying the effect of different types of seats, the patients successively sat on a stool (height=46 cm), a chair (height=47 cm), an office chair (height=47 cm), a bench (height=66 cm), a physiotherapy ball (pezzì ball, diameter=70 cm), and a knee stool (balance). The patients also sat on a stool with a padded wedge (angle=9.5°). Sitting on a chair and an office chair, respectively, was measured with and without contact to the backrest. The effect of the arm position was also studied: the arms were either hanging down laterally or the hands or arms were placed on the thighs or on the armrests.

Evaluation

Only the maximum resultant forces (geometrical sum of the three force components) on the implant are reported here. The measured moments were small during sitting. The different parameters studied were measured in up to four measuring sessions. Besides the posture, the loads on the implants also depend on the increasing stiffness of the added bone material and thus on the postoperative time. Therefore, relative values were determined, and comparisons were made only when the measurements were performed on the same day. Usually, the median values are presented here. It was merely descriptive statistics, which could be carried out, because no more than five patients had a telemeterized spinal implant.

Results

When sitting on a stool, the inclination of the upper body had a strong effect on the resultant implant force (Fig. 2).

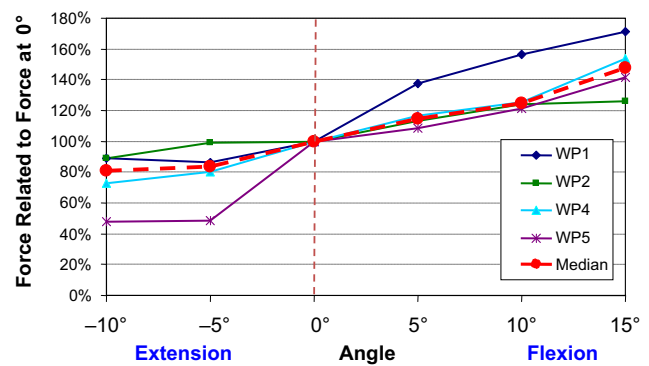


Fig. 2. Effect of inclination angle of the upper body on implant loads when sitting on a stool. Median values for each of the four patients and the median values of these medians are given.

Flexion led to a force increase, whereas extension reduced the implant force. The average resultant force increase for 15° flexion was 48%, and the force decrease for 10° extension was 19%.

The force on the VBR decreased with an increasing declination angle of the backrest for each subject when sitting on a chair with an adjustable backrest (Fig. 3). There were great interindividual differences in load levels and their decrease. The largest load reduction occurred between 90° and 130°. In three patients, there was already a load reduction of approximately 60% when leaning against a slightly declined backrest, whereas in the other two patients, the load reduction at small declination angles was much less. In one patient (WP2), there was a nearly linear reduction up to a backrest angle of 160°. The values for an angle of 180° were between 17% and 45% of those for sitting relaxed and equivalent to those for lying in a supine position.

The median value of the resultant implant force tends to increase with increasing seat height (Fig. 4). The average force difference between the seat height 40 and 60 cm was 17% in relation to the value for a seat height of 50 cm.

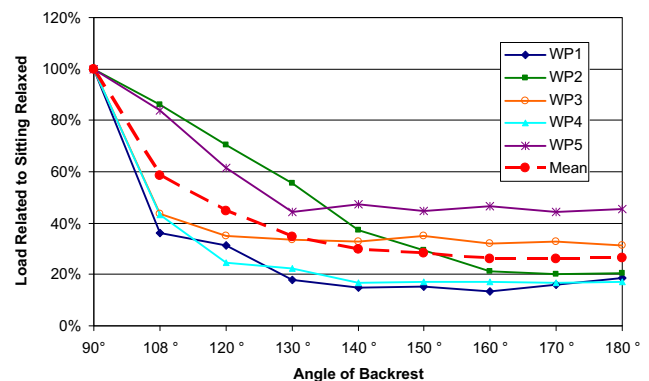


Fig. 3. Influence of backrest inclination on implant forces when sitting on a chair with an adjustable backrest as measured in five patients. Median values for each of the five patients and the mean values of these medians are shown.

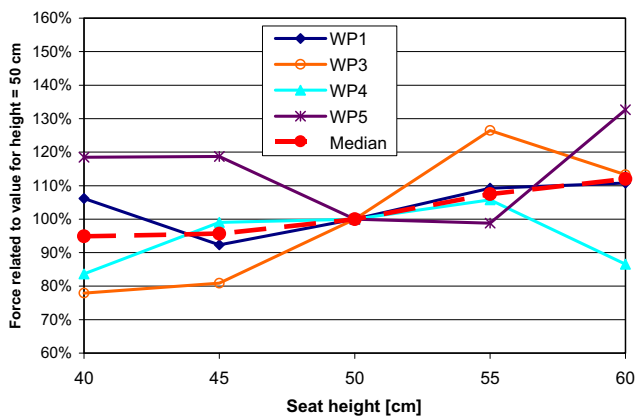


Fig. 4. Effect of seat height on forces on the vertebral body replacement. Median values for each of the four patients and the median values of these medians are shown.

However, this average trend was not present in any individual data.

In comparison to sitting on a stool, the average load reduction was 35% when sitting on a chair, 41% when sitting on an office chair, 7% when sitting on a bench, and 19% when using a knee stool (Fig. 5). The average force was increased by 7% when sitting on a physiotherapy ball. A padded wedge on a stool reduced the average load by 9%. The ranges here, however, are large, especially for the chair and office chair. When the patient's back had no contact with the backrest, the load reductions were only 9% and 19%, respectively, for sitting on a chair and an office chair.

Leaning against the backrest of a chair reduced the implant loads on the average by 36%. Placing the hands on the thighs reduced the implant load of the five patients on the average by 19% compared with the posture where the arms were hanging laterally. The load reduction was even higher when the arms were placed on the armrests. The position of the head had also an influence on the implant loads. A forward-flexed head shifts the center of gravity of the upper body anteriorly and thus leads to higher spinal loads (Fig. 6).

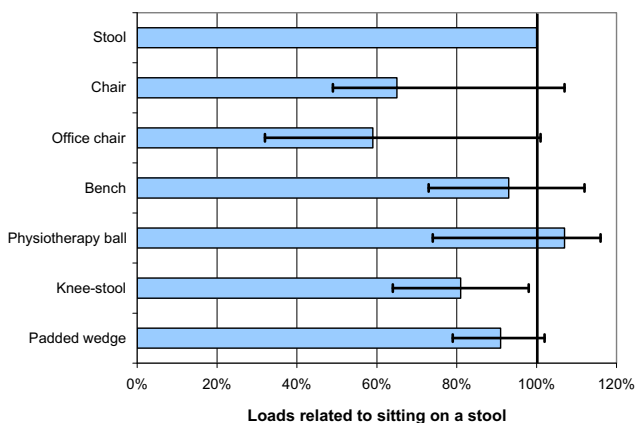


Fig. 5. Normalized forces on the vertebral body replacement for sitting on different types of seats. Sitting on a stool was set to 100%.

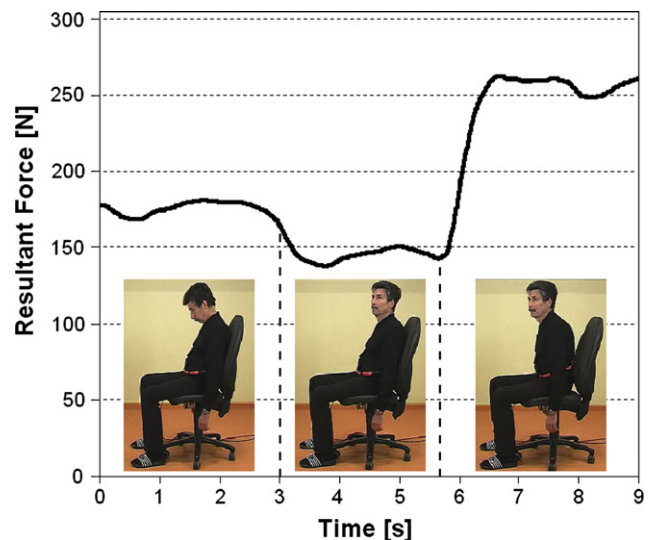


Fig. 6. Measured forces while sitting on a chair. At the beginning, the patient was leaning against the backrest with the head flexed. After about 3 seconds, the head was moved into a neutral position. Because of this head movement, the resultant force was reduced. After about 5.6 seconds, the patient moved the upper body forward, losing contact with the backrest. This increased the resultant force.

Discussion

The loads on a telemeterized VBR were measured to determine the effects of several parameters, including inclination of the upper body, declination of the backrest, seat height, and type of seat.

The following limitations of the study should be mentioned: The telemeterized implants have been implanted in a small cohort of only five patients and thus only descriptive statistics could be applied. For the measurements done here, a power coil had to be placed around the patient's trunk. Some patients felt uncomfortable when leaning against a backrest. This may affect the measured loads.

At the implant level, the spinal load is shared between the VBR, the remaining part of the vertebra, the added bone material, and the internal spinal fixation device. Thus, the measured load represents only one part of the spinal load. The VBRs were implanted at different levels. The absolute values of the spinal load depend on the spinal level and on the body weight of the patients. For these reasons, only measurements performed on the same day are compared, and the medians of relative values are presented. We assume that the relative changes of the implant loads correlate strongly with the relative changes in spinal loads.

Only short instructions were given to the patients for how to take on a certain posture on a seat. For example, they were not explicitly told exactly how to sit on the seat, where to place the feet, how to hold the head, and so on. In this way, we hoped to measure typical postures of daily living and avoid unnatural restrictions. Because of this, there is a great variation in the postures and in the measured loads. Precise instructions would have probably reduced

the variation of the results. However, to follow such instructions, it takes much more time, is more stressful for the patients (they are older than 60 years), may cause spasm of the back muscles and lead to higher unrealistic loads, so that the patients might therefore refuse further measurements. If patients get a new task, especially if it is hitherto unknown for them, muscles are activated and joint loads increase. Therefore, it is very important that the patients feel comfortable during the measurements. This is not the case when they have to repeat a task several times because they did not follow exactly the instructions.

Forward or backward inclination of the upper body changes the center of gravity of the upper body, and thus the dorsal muscle forces required to balance the upper body. Higher trunk muscle forces are accompanied by higher spinal forces. The measured effect of the center of gravity position is in agreement with the intradiscal pressure measurements of Wilke et al. [7,8].

With an increasing declination angle, the backrest of a chair with an adjustable backrest carries an increasing part of the upper body, which reduces the trunk muscle forces required for keeping the upper body in the desired position. A backrest angle of 180° corresponds to lying in a supine position. For small backrest declination angles, the load reduction differed strongly. The load reduction is probably influenced by the friction between the patient's clothes and backrest, and by the degree of relaxation of the patient.

With an increasing seat height, the average measured force value increased. There is probably a tendency to bend the upper body a little bit forward with increasing seat height and reduce the lordosis of the lumbar spine. The large variation of the curves for the different patients indicates that the sitting posture also varied strongly between the different seat heights. One has also to keep in mind that the absolute magnitude for the reference value differed considerably between different patients and different measuring sessions. We do not recommend using a chair with low seat height because more effort is usually required for rising from a seat with a low height, compared with that of a high one.

For all types of seats, except the physiotherapy ball, the average measured loads were below those when sitting on a stool. The bending moment measured with telemeterized fixation devices were in all cases less than that for sitting on a stool [12]. However, the load differences were often more pronounced in the present study. In the fixator study [12], the patients were not allowed to lean against the backrest when sitting on a chair or office chair.

Sitting on a chair or an office chair led to a strong reduction of the implant load when the patient's back had contact to the backrest. Without contact, the load reduction was only about 10% to 20% compared with sitting on a stool. This demonstrates that the seat itself has only a small effect on the resultant force. The inclination of the upper body (load increase of about 48% for 15°

flexion), contact with the backrest (load reduction 36%), and the position of the arms (reduction of about 19%) are more important.

Slightly higher implant loads were measured for sitting on a physiotherapy ball than on a stool. The intradiscal pressure measured by Wilke et al. [8] also was higher when sitting on a physiotherapy ball than on a stool. A ball is not as stable as a stool. Thus, additional muscle forces are necessary to balance the upper body while sitting on a ball. Generally, higher muscle forces are accompanied by higher joint and thus spinal forces. In addition, especially elderly people often feel unsafe when sitting on a ball. This may lead to a slightly flexed upper body, which also leads to higher spinal loads.

Sitting on a knee stool caused lower implant loads than sitting on a stool. At least at the beginning, the upper body is kept erect while sitting on a knee stool. For our measurements, the patients used the knee stool only for a short period (less than 5 minutes). It is not known how the loads will change in the course of time.

We assume that the spinal loads during sitting are not harmful to the spine as long as the upper body position is varied from time to time. For monotone postures, however, even small loads may already cause low back pain when sitting in the same position for a long period. Leaning against the backrest reduces the spinal loads and thus allows sitting for a longer time without fatigue. To loll on a seat usually reduces the spinal loads, especially when the back reclines. Changing the position during sitting is very important and should be performed regularly.

Conclusions

Leaning against a backrest while sitting reduces the spinal loads considerably. The spinal loads are also reduced by placing the arms or hands on the thighs or even on the armrests. On the contrary, forward inclination of the head or upper body leads to higher loads. In general, moving the center of gravity of the upper body in the anterior direction increases the spinal loads, whereas supporting the upper body decreases them. These are the main factors influencing the spinal loads. The type of seat is a secondary factor and affects the loads only when it is also the corresponding primary factor, which is varied.

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